

Concepts (TIL-ACT versus CART)

TIL - Tumor-Infiltrating Lymphocyte

ACT - Adoptive Cell Therapy

TIL-ACT stands for Tumor-Infiltrating Lymphocyte Adoptive Cell Therapy. It is a personalized cancer immunotherapy that extracts a patient's own immune cells from a tumor, multiplies them in a lab to number in the billions, and infuses them back into the body to attack the cancer.

- **Tumor collection:** A surgeon removes a piece of the patient's tumor tissue. [What is TIL therapy for cancer and how does it work? | Ohio State Medical Center](#)
- **Cell extraction and growth:** Technicians isolate the lymphocytes living inside the tumor and grow them in a lab using growth factors like IL-2. [Adoptive cell therapy with tumor-infiltrating lymphocytes in combination with nivolumab in patients with advanced melanoma, Tumor-infiltrating lymphocyte \(TIL\) therapy](#)
- **Lymphodepletion:** The patient receives a short round of chemotherapy to clear space in the immune system for the new cells. [Tumor Infiltrating Lymphocyte \(TIL\) Therapy for Solid Tumor Treatment: Progressions and Challenges, Tumor-Infiltrating Lymphocyte Therapy \(TIL Therapy\): Amplifying Your Body's Cancer-Fighting Cells - youtube](#)
- **Cell infusion:** Billions of the supercharged T-cells are returned to the patient's bloodstream to seek out and destroy tumors. [What is TIL Therapy? - youtube - Dana_Farber](#) and [Tumor-Infiltrating Lymphocyte Therapy \(TIL Therapy\): Amplifying Your Body's Cancer-Fighting Cells - youtube - UChicago](#)
- **Immune stimulation:** Additional doses of IL-2 are given after the infusion to help the cells multiply and persist inside the body. [Optimizing TIL therapy for uveal melanoma: lessons learned and unlearned from cutaneous melanoma](#) and [Adoptive cell therapy with tumor-infiltrating lymphocytes in combination with nivolumab in patients with advanced melanoma](#)

Key Uses

- Primarily studied and applied in solid tumors like metastatic melanoma.
- Expanding into clinical trials for other solid tumors like cervical cancer and lung cancer. [PragmaTIL Receives Approval to Advance Safer Cell Therapy for Cancer Patients](#) and [Achievements and challenges of adoptive T cell therapy with tumor-infiltrating or blood-derived lymphocytes for metastatic melanoma: what is needed to achieve standard of care?](#)

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what are the differences between CART-Cell treatment and TIL-ACT?

Both are adoptive cell therapies — lymphodepletion followed by infusion of a large ex vivo–expanded autologous T-cell product — but they differ in where the cells come from, how they recognise the tumour, and consequently what they fail against.

	TIL-ACT	CAR-T
Cell source	Lymphocytes resident in a resected tumour lesion	Peripheral blood T cells (leukapheresis)
Recognition	Native, unmodified TCRs → peptide–HLA complexes	Synthetic receptor: scFv ectodomain + hinge/TM + costim (CD28 or 4-1BB) + CD3ζ → surface antigen, HLA-independent
Specificity breadth	Polyclonal, many antigens (neoantigens, cancer-germline, differentiation antigens); specificity is unknown at infusion	Monospecific, one defined antigen chosen in advance
Genetic modification	None (product is selected/expanded, not engineered)	Viral transduction or non-viral integration; requires vector manufacture, VCN and replication-competent-vector release testing
Manufacturing	Tumour fragments → pre-REP outgrowth in high-dose IL-2 → rapid expansion with anti-CD3 and feeders; ~4–6 weeks, 10^9 – 10^{11} cells	Selection, activation, transduction, expansion; ~1–3 weeks to product release
Prerequisite	A resectable lesion that yields viable, expandable TIL	Adequate circulating T cells; antigen expressed on the tumour
Post-infusion support	Classically high-dose bolus IL-2	None
Dominant toxicities	IL-2 toxicity (capillary leak, hypotension), lymphodepletion cytopenias; CRS uncommon and usually mild	CRS, ICANS, prolonged cytopenias, on-target/off-tumour injury, lineage aplasia (e.g. B-cell with CD19)
Where it works	Solid tumours — lifileucel received FDA accelerated approval in 2024 for unresectable/metastatic melanoma after anti-PD-1, the first TIL product licensed	B-cell malignancies and myeloma (CD19, BCMA); solid-tumour CAR-T remains investigational

The failure modes are mirror images. CAR-T is blind to HLA, so $\beta 2M$ or HLA class I loss — a common melanoma immune-escape route — does not disable it; but antigen loss or downregulation does, and a single surface target must be near-uniformly expressed and absent from vital normal tissue, which is why solid tumours with heterogeneous, mostly intracellular antigens have resisted the approach. TIL recognises intracellular and neoantigen-derived peptides and hits many targets at once, so heterogeneity is less fatal, but every clone depends on intact antigen presentation, and the product's potency is not defined in advance — it is whatever tumour-reactive clonotypes happened to grow out.

TCR-T sits between them: blood-derived T cells engineered with a native-format TCR against a defined peptide–HLA pair (e.g. MAGE-A4 in synovial sarcoma). HLA-restricted like TIL, defined and manufacturable like CAR-T, and restricted to patients carrying the right HLA allele.

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Response-arm filter

I checked two sources per study:

1. the per-sample GEO characteristics (scanned all 52 tumour-tissue series for response/RECIST/outcome fields) and
2. the abstracts of the linked publications.

No study in the shortlist carries response labels in its GEO metadata.

Across all 52 tumour-tissue series, exactly one does — GSE335480 (TIL-ACT, clinical response: PR / SD / PD per sample) — and its raw reads are not public, so it fails the scVelo criterion. For everything else, responder status has to be lifted from the paper and joined on the patient ID in the sample titles.

5 of 15 have both responders and non-responders profiled, across three cohorts:

GEO	Therapy	Timepoints	Velocity on	Raw reads	PMID
GSE269936	anti-PD-1 ± anti-CTLA-4	single, mostly pre-ICI	malignant + immune	179 runs, 1.68 Tb	40890106
GSE222446	TIL-ACT phase 1 (prior ICI)	pre-ACT (T0), 10 patients	malignant + immune	13 runs, 0.22 Tb	38306416
GSE221553	TIL-ACT phase 1 (same cohort)	pre- and post-ACT,	13 patients	CD45+ immune only 22 runs, 0.29 Tb	38306416
GSE229858	TIL-ACT	ACT product, 13 patients	T cells only	13 runs, 0.17 Tb	39276771
GSE261023	TIL-ACT	baseline tumour + product, 2 patients	T cells (multiome)	3 runs, 0.06 Tb	39276771

The **two TIL-ACT papers state the responder/non-responder contrast directly** — the Science Immunology study reports

1. higher basal tumour-intrinsic immunogenicity and
2. mutational burden in ACT responders with distinct CD8 TIL and myeloid states, and
3. the Immunity study tracks clonotypes from baseline tumour through product to post-ACT blood and tumour,
4. contrasting patients with and without clinical response. GSE269936 stratifies on PFS under ICI rather than RECIST categories, so its "+/-" axis is time-to-progression, not a binary label.

Two partial matches: GSE174401 (mixed systemic therapy, stratified by overall survival with one long-term survivor against poor survivors) and GSE139249 (NK cells including lesions from patients who had progressed on checkpoint blockade — progressors only, no responder arm). The remaining 8 are treatment-naïve cohorts, atlases, a

radiosurgery/dexamethasone trial (GSE275929 — its two arms are steroid dose, not immunotherapy response), or ex vivo ICB modelling on organoids (GSE111360).

If the goal is velocity in malignant cells with a response contrast, the field narrows to GSE269936 and GSE222446. For the Lausanne cohort, GSE222446 (unsorted 10x 3' GEX, pre-ACT) and GSE221553 (CD45+ 5' GEX, pre + post-ACT) are the same 13 patients, with patient1...patient13 IDs in both the titles and the paper's supplementary tables, so responder labels transfer cleanly and the two can be analysed as one cohort.

One correction to the previous shortlist: GSE222750 is scTCR-seq only (cellranger vdj, contig annotation files, no gene-expression libraries) — its GEX counterpart is GSE222446. It is now moved to the exclusions, so the scVelo-ready set is 15 studies, not 16.

1. melanoma_scvelo_immunotherapy_response.csv — all 15 studies with therapy, response-arm verdict, the evidence sentence behind each call, where the labels live, timepoints and run counts
2. melanoma_scvelo_ready_studies.csv — corrected 15-study shortlist (v2)
3. melanoma_scvelo_excluded_with_reasons.csv — now 37 rows including GSE222750 (v2)